

Mathematics of Neural Network Back-propagation

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How much do w11 contribute to L?

$$\begin{aligned} H21 &= h11.wh11 + h12.wh21 \\ \delta h21/\delta wh11 &= h11 \\ \delta h21/\delta h12 &= wh21 \\ \\ H12 &= w11.x1 + w13.x2 \\ \delta h12/\delta w11 &= x1 \\ \\ \delta L/\delta w11 &= \delta L/\delta h12 * \delta h12/\delta w11 \\ &= \delta L/\delta h12 * x1 \\ \\ \delta L/\delta h12 &= (\delta L/\delta h21 * \delta h21/\delta h12) + \\ &\quad (\delta L/\delta h22 * \delta h22/\delta h12) \\ \\ \delta h22/\delta h12 &= wh22 \\ \\ \delta L/\delta h21 &= \delta L/\delta pred * \delta pred/\delta h21 \\ &= \delta L/\delta pred * w1 \\ \delta L/\delta h22 &= \delta L/\delta pred * w2 \\ \\ \delta L/\delta h12 &= \delta L/\delta pred * w1 * wh21 + \delta L/\delta pred * w2 * wh22 \\ \\ \delta L/\delta w11 &= (\delta L/\delta pred * w1 * wh21 + \delta L/\delta pred * w2 * wh22) * x1 \end{aligned}$$

How much do w10 contribute to L?

$$\begin{aligned} H11 &= w10.x1 + w12.x2 \\ \delta h11/\delta w10 &= x1 \\ h21 &= h11.wh11 + h12.wh21 \\ \delta h21/\delta h11 &= wh11 \\ \\ H22 &= wh12.h11 + wh22.h12 \\ \delta h22/\delta h11 &= wh12 \\ \\ \delta L/\delta w10 &= \delta L/\delta h11 * \delta h11/\delta w10 \\ \\ \delta L/\delta h11 &= (\delta L/\delta h21 * \delta h21/\delta h11) + (\delta L/\delta h22 * \delta h22/\delta h11) \\ \\ \delta L/\delta h21 &= \delta L/\delta pred * w1 \\ \delta L/\delta h22 &= \delta L/\delta pred * w2 \\ \\ \delta L/\delta h11 &= (\delta L/\delta pred * w1 * wh11) + (\delta L/\delta pred * w2 * wh12) \\ \\ \delta L/\delta w10 &= ((\delta L/\delta pred * w1 * wh11) + (\delta L/\delta pred * w2 * wh12)) * x1 \end{aligned}$$

How much did wh12 contribute to L?

$$\begin{aligned} H22 &= h11.wh12 + h12.wh22. \\ \delta h22/\delta wh12 &= h11 \\ \delta L/\delta h22 &= \delta L/\delta pred * \delta pred/\delta h22 \\ \delta pred/\delta h22 &= w2 \\ \\ \delta L/\delta wh12 &= \delta L/\delta h22 * \delta h22/\delta wh12 \\ &= \delta L/\delta pred * w2 * h11 \\ &= -0.995 * 0.02 * 0.32 \end{aligned}$$

How much did wh11 contribute to L?

$$\begin{aligned} H21 &= h11.wh11 + h12.wh21. \\ \delta h21/\delta wh11 &= h11 \\ \delta L/\delta h21 &= \delta L/\delta pred * \delta pred/\delta h21 \\ \delta pred/\delta h21 &= w1 \\ \\ \delta L/\delta wh11 &= \delta L/\delta h21 * \delta h21/\delta wh11 \\ &= \delta L/\delta pred * w1 * h11 \\ &= -0.995 * 0.02 * 0.32 \end{aligned}$$

Error at h21=**how much did w1 contribute to L?**

$$\begin{aligned} \delta L/\delta w1 &= \delta L/\delta pred * \delta pred/\delta w1 \\ \delta pred/\delta w1 &= h21 \\ &= -0.0995 * 0.0584 \end{aligned}$$

$$L = 1/2 (exp - pred)^2$$

$$\begin{aligned} u &= exp - pred \\ L &= 1/2 (u)^2 \\ \delta u/\delta pred &= -1 \\ \delta L/\delta u &= u \\ \delta L/\delta pred &= \delta u/\delta pred * \delta L/\delta u \\ &= -1 * u \\ &= - (exp - pred) \\ &= -exp + pred \\ &= pred - exp \end{aligned}$$

$$pred = 0.0053, exp=1$$

$$pred = w1.h21 + w2.h22 + b$$

$$\begin{aligned} \text{error at o} &= \delta L/\delta pred \\ &= (pred - exp) \\ &= 0.0053 - 1 = -0.995 \end{aligned}$$

We will share -0.995 backwards

Error at h22:

how did w2 contribute to L?

$$\begin{aligned} \delta L/\delta w2 &= \delta L/\delta pred * \delta pred/\delta w2 \\ \delta pred/\delta w2 &= h22 \\ &= -0.995 * 0.218 \end{aligned}$$

How much did wh21 contribute to L?

$$\begin{aligned} \delta L/\delta wh21 &= \delta L/\delta h21 * \delta h21/\delta wh21 \\ \delta L/\delta h21 &= \delta L/\delta pred * \delta pred/\delta h21 \\ \delta pred/\delta h21 &= w1 \\ H21 &= h11.wh11 + h12.wh21 \\ \delta h21/\delta wh21 &= h12 \\ \\ \delta L/\delta wh21 &= \delta L/\delta pred * \delta pred/\delta h21 * h12 \\ &= -0.995 * 0.02 * 0.26 \end{aligned}$$

How much did wh22 contribute to L?

$$\begin{aligned} \delta L/\delta wh22 &= \delta L/\delta h22 * \delta h22/\delta wh22 \\ \delta L/\delta h22 &= \delta L/\delta pred * \delta pred/\delta h22 \\ \delta pred/\delta h22 &= w2 \\ H22 &= h11.wh12 + h12.wh22 \\ \delta h22/\delta wh22 &= h12 \\ \\ \delta L/\delta wh22 &= \delta L/\delta pred * \delta pred/\delta h22 * h12 \\ &= -0.995 * 0.02 * 0.26 \end{aligned}$$

The δL w.r.t to each weight is called the gradient. These values is what will be used to update the weights. For e.g, $\delta L/\delta w22$ will be used to update w22.

$$W22 = w22 - \eta \delta L/\delta w22$$

How much do w13 contribute to L?:

$$\begin{aligned} H12 &= w13.x2 + x1.w11 \\ h22 &= h12.wh22 + h11.wh12 \\ \delta L/\delta w13 &= \delta L/\delta h12 * \delta h12/\delta w13 \\ \\ \delta h12/\delta w13 &= x2 \\ \delta L/\delta h12 &= \delta L/\delta pred * \delta pred/\delta h22 * \delta h22/\delta h12 + \delta L/\delta pred * \delta pred/\delta h21 * \delta h21/\delta h12 \\ \\ \delta h22/\delta h12 &= wh22 \\ \delta h21/\delta h12 &= wh21 \\ \\ \delta L/\delta h12 &= \delta L/\delta pred * \delta pred/\delta h22 * wh22 + \delta L/\delta pred * \delta pred/\delta h21 * wh21 \\ \\ \delta L/\delta w13 &= (\delta L/\delta pred * \delta pred/\delta h22 * wh22 + \delta L/\delta pred * \delta pred/\delta h21 * wh21) * x2 \\ &= ((\delta L/\delta pred * w2 * wh22) + (\delta L/\delta pred * w1 * wh21)) * x2 \end{aligned}$$

How much do w12 contribute to L?

$$\begin{aligned} H11 &= w12.x2 + w10.x1 \\ \delta h11/\delta w12 &= x2 \\ H21 &= h11.wh11 + wh21.h12 \\ \delta h21/\delta h11 &= wh11 \\ H22 &= h12.wh22 + h11.wh12 \\ \delta h22/\delta h11 &= wh12 \\ \\ \delta L/\delta w12 &= \delta L/\delta h11 * \delta h11/\delta w12 \\ \\ \delta L/\delta h11 &= \delta L/\delta h21 * \delta h21/\delta h11 + \delta L/\delta h22 * \delta h22/\delta h11 \\ &= \delta L/\delta pred * w1 * wh11 + \delta L/\delta pred * w2 * wh12 \\ \\ \delta L/\delta w12 &= ((\delta L/\delta pred * w1 * wh11) + (\delta L/\delta pred * w2 * wh12)) * x2 \end{aligned}$$

